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Who Guards the Guardians? The Challenges of Evaluating Identifiability of Learned Representations

(/pdf?id=8rxSN6cNtv)

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Keywords: identifiable representation learning, causal representation learning, disentangled representation learning, unsupervised learning

TL;DR: All existing identifiability metrics can be deceptive; we characterise when they fail.

Abstract:
Identifiability in representation learning is commonly evaluated using standard metrics (e.g., *MCC*, R^2, *DCI*) on synthetic benchmarks with known ground-truth factors. These metrics are assumed to reflect recovery up to the equivalence class guaranteed by identifiability theory. We show that this assumption holds only under specific structural conditions: each metric implicitly encodes assumptions about both the data-generating process (DGP) and the encoder. When these assumptions are violated, metrics become misspecified and can produce systematic false positives and false negatives. Such failures occur both within classical identifiability regimes and in post-hoc settings where identifiability is most needed. We introduce a taxonomy separating DGP assumptions from encoder geometry, use it to characterise the validity domains of existing metrics, and release an evaluation suite for reproducible stress testing and comparison.

Submission Number: 1105

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Official Review of Submission1105 by Reviewer YTF7

Official Review by Reviewer YTF7 09 Apr 2026 at 15:47 (modified: 24 Apr 2026 at 04:15)
Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer YTF7, Authors Revisions (/revisions?id=gIVtL6p7fe)

Q1 summary:
The paper argues that standard identifiability metrics encode implicit structural assumptions whose violation yields population-level false positives and negatives at n → ∞. It builds a factor-structure × encoder-geometry taxonomy, formalizes four desiderata P1–P4 (ρ-invariance, d_eff-sensitivity, OC-invariance, uninformative-sensitivity), proves no metric satisfies all four, and gives a closed-form MCC failure under D_ρ + E3 (§F.1) plus a null-encoder bound E[MCC] ≥ √(2 log(m/n)) (§F.3).

Q2 novelty: 3: Good: The paper makes non-trivial advances over the current state-of-the-art.
Q2 correctness: 3: Good: The paper appears to be technically sound, but I have not carefully checked the details.
Q2 evidence based support: 3: Good: the main claims are supported by convincing evidence (adequate experimental evaluation, proofs, code, references, assumptions).
Q2 reproducibility: 3: Good: key resources are available and key details are sufficiently well-described for competent researchers to confidently reproduce the main results.
Q2 writing clarity: 3: Good: The paper is well organized but the presentation could be improved.
Q3 strengths:
The paper's central move — framing metric failure as *population-level misspecification* rather than finite-sample noise or algorithm-specific inductive bias — is the right framing, and as far as I know it is new. Locatello et al. (2019) showed that unsupervised disentanglement requires inductive bias in the *learner*; the present paper is orthogonal: it shows that even supervised identifiability metrics are structurally misspecified under conditions the identifiability theorems explicitly permit. The two-axis taxonomy (latent factor structure × encoder geometry) organizes this failure mode systematically.

The theory–experiment coupling is tight. §3.1 and §F.1 derive MCC's closed form under D_ρ + E3 (Eq. (18)–(44)) and show that as ρ → ±1 the MCC approaches 1 even for a fully entangled encoder; Figure 2 makes this analytical prediction visible. §F.3 derives E[MCC] ≥ √(2 log(m/n)) for a null encoder via a CLT-plus-Hungarian-matching argument, and Figure 5's heatmap independently verifies that m/n, not m/d, controls the false-positive inflation. I found this second result the most striking in the paper: it identifies an evaluation-setting variable (the representation-to-sample ratio) that is routinely exceeded in current mechanistic-interpretability work, and the analytical prediction is testable in one line.

The practical output is unusual for a formal paper. Appendix A provides a 10-item practitioner checklist; Tab 3 is a metric × property truth table; Tab 1 gives explicit numerical thresholds for m/n and m/d. Appendix C is a systematic review of 62 CRL and nonlinear ICA papers from 2020–2025 across NeurIPS, ICLR, ICML, AISTATS, UAI, AAAI, CLeAR, and JMLR, reporting that 25 used MCC, 9 used R^2, and only 2 used DCI — this quantifies how widespread the misuse pattern is.

The encoder taxonomy covers every dimension-ratio regime. E1 through E9 span m = d (perfect disentangled, elementwise nonlinear, linearly entangled), m < d (undercomplete), and m > d (four distinct overcomplete geometries plus the null baseline). Each comes with an analytic construction, which makes the stress-test suite reproducible without access to trained encoders. The running example in §2 — a resistor circuit governed by R = R_0(1 + α T) and V = IR — grounds the abstract D_f / d_eff concepts in physical intuition.

Q4 weaknesses:
Figure 1's caption contradicts Tab 3 and Fig 4 on R^2's P3 status. Figure 1 caption (p. 2) says "Only R^2 is robust across (P1), (P3), and (P4), but shares the (P2) limitation." Tab 3 (p. 8) marks R^2 × P3 as ✗. Fig 4 (p. 8) states "R^2 collapses for nonlinear E6," which is precisely a P3 failure. Two locations say R^2 fails P3; the headline caption says it does not. "R^2 is the only metric robust across P1, P3, P4" is one of the paper's three takeaway sentences; if it is wrong, the story must be rewritten as "no single metric satisfies P1, P3, P4 jointly" — a strictly weaker positioning. A related inconsistency: §3.3 notes that "MCC cannot be used for distributed codes (E8)" — a second, distinct MCC failure mode under overcompleteness not flagged in Tab 3's MCC row, which uses a single ✗ without distinguishing mechanisms. The authors must either clarify that Tab 3's ✗ covers both mechanisms or split the row.

The experimental evidence uses only analytically constructed encoders. The limitations acknowledge this, but the gap between "metric fails under this analytical construction" and "metric fails on encoders people actually train" is the question UAI reviewers will ask first. Anthropic's published Llama-3 and Claude-Sonnet sparse autoencoders have $m \approx 65 \times 10^3$ features and n typically under 10^4 probing samples, giving $m/n \geq 6.5$. §3.4's formula then bounds the null-encoder expected MCC from below by $\sqrt{2 \log 6.5} \approx 2.16$, which saturates the metric. The MCC-type numbers in recent SAE interpretability work may therefore be statistical floor effects rather than evidence of identifiability — a single paragraph of §3.4 applied to Anthropic's reported numbers would make this concrete and would change how the interpretability community reports metrics.

MIG, InfoMEC, and T-MEX appear only in Tab 3 and §G. Tab 3 lists a P1–P4 truth table for six metrics, but §3 walks through only MCC, DCI-D, and R^2. MIG (Chen et al. 2018), InfoMEC (Hsu et al. 2023), and T-MEX (Yao et al. 2025) are never visualized in the main text. Two of these are 2023+ metrics rising quickly in SAE evaluation; burying their failure cases in the appendix leaves "no metric works" with half its legs standing behind the main text. At minimum, one failure case per metric should appear in Figure 1 or a successor figure.

Q5 comments:

Required experiments for rebuttal

Any of the following three would move the score from 6 to 8. All are executable during the rebuttal period and have falsifiable outcomes tied to specific weaknesses above.

1. Anthropic SAE case study. For m = 65 536 and $n \in \{500, 1000, 5000, 10\,000\}$, the §3.4 formula predicts null-encoder expected MCC bounds of approximately 2.16, 1.92, 1.52, and 1.27 respectively (the first two saturate the metric at 1). Request Anthropic's published MCC numbers for at least two n values within this range and report the comparison. If the published MCCs sit at or near the null-encoder bound, the paper has identified a live problem in a high-profile interpretability paper; if they sit well below, the paper gains a calibration data point. Either outcome is publishable evidence.
2. Locatello et al. (2019) pretrained-model application. Select the top-10% and bottom-10% MCC scores from the 7 000 pretrained models in Locatello et al. (2019) (50 models each), apply the Tab 1 m/n correction threshold, and re-report MCC. If the top-10% average drops by ≥ 20 % while the bottom-10% average is unaffected, the checklist is empirically validated; if not, Tab 1's thresholds need re-examination. The Locatello models are publicly available, which makes this executable within the rebuttal window.
3. Resolve the Fig 1 / Tab 3 / Fig 4 inconsistency. Either correct the Figure 1 caption to read "Only R^2 is robust across (P1) and (P4); all metrics fail on (P3) for some encoder geometry," or revise the Tab 3 R^2 × P3 entry from ✗ to ~ with an explicit threshold definition. The internal contradiction must not survive to camera-ready.

Technical issues

The $\rho \to 1$ limit of Eq. (39) gives a clean closed-form result that belongs in the main text. When $\rho \to 1$, the numerator $1 + \varepsilon\rho \to 1 + \varepsilon$ and the denominator $\sqrt{1 + \varepsilon^2 + 2\varepsilon\rho} \to \sqrt{(1 + \varepsilon)^2} = 1 + \varepsilon$, so MCC $\to (1/3)(1 + 2) = 1$. Moving this one-line computation into §3.1 strengthens the "MCC saturates at 1 for an entangled encoder as $\rho \to 1$" claim without cost.

§F.3's $\sqrt{2 \log(m/n)}$ bound uses Cai & Jiang (2011) Theorem 2. Hungarian matching on best-of-m picks introduces an additional multiplicative factor that the text glosses over; please state the exact constant and cite the relevant lemma. §F.3 should also state the continuation convention for $m/n < 1$.

P1 and P4 interact under correlated factors. The null-encoder experiments assume ρ and m/n vary independently, but when the ground-truth factors are highly correlated, the effective dimension drops, which changes the constant in $\sqrt{2 \log(m/n)}$. A sentence addressing this coupling would make the result robust.

Definition 2 defines d_eff as $d - k$ (number of independent constraints), but under multiple or non-independent constraints this is not well-defined. A sentence clarifying the intended scope would help.

Minor writing issues

On p. 2 the text reads "MCC scaling in the order of $\sqrt{2 \log m/n}$"; for consistency with §F.3 this should be "MCC scales as $\sqrt{2 \log(m/n)}$." The Figure 1 P3 panel caption should state that the ground-truth identifiability holds for the chosen encoder family throughout the m/d sweep, so that the upward MCC trend cannot be misread as a real identifiability change. Tab 3's caption should define ✓ / ~ / ✗.

Additional concerns

The finite-sample-to-asymptotic transition in §3.4 lacks quantitative calibration. The text states that the $\sqrt{2 \log(m/n)}$ behavior dominates for $m/n \geq 0.1$, but no table gives the theory-vs-empirics relative error at specific m/n points. A second issue: for $m/n < 1$ the quantity $\log(m/n)$ is negative, so $\sqrt{2 \log(m/n)}$ is undefined — yet Figure 5 sweeps m/n down to 0.01. §F.3 should explain the continuation convention (presumably the bound becomes trivial for $m/n < 1$ and the observed inflation comes from finite-sample effects only).

The "no metric satisfies P1–P4" negative result has no impossibility lower bound. The paper shows that existing metrics all fail but does not prove or conjecture that any metric satisfying all four properties must use higher-order moments, must know d_eff, or must include a null-encoder calibration step. A single conjectural impossibility — even a sketch — would convert this from a negative survey into a structural limit and give metric designers concrete guidance.

R^2's collapse on nonlinear E6 has no mechanistic explanation. The paper positions R^2 as "the most robust metric available," yet Fig 4 shows it collapsing under nonlinear overcomplete encoders. §3.3 describes the collapse in one sentence with no algebra. §F should derive the closed-form or asymptotic expression for R^2 under E6, in parallel to what §F.1 does for MCC under E3.

The §C 62-paper survey lacks methodological transparency. The Semantic Scholar API call is mentioned, but not the query strings, inclusion/exclusion criteria, de-duplication rules, or inter-rater reliability. The 25/62 MCC-usage number is the headline that justifies the paper's motivation, so its methodology must be reproducible at the level a systematic review requires.

Tab 2's caption uses "[CITE]" as an unfilled BibTeX placeholder for the MIG entry; this should be Chen et al. (2018). Tab 3 uses ✓ / ~ / ✗ without defining the thresholds; the caption should read, for example, "✓ = ≥ 95 % of tested settings, ~ = 50–95 %, ✗ = < 50 %." Code release is deferred to acceptance; OpenReview supports anonymous supplementary material at submission time, so the current stance reduces reproducibility without necessity.

Questions for the authors

1. Does R^2 satisfy P3 or not? The Figure 1 caption says yes; Tab 3 and Fig 4 say no. Please give a single answer and correct the inconsistent location.
2. Can P1–P4 be merged into a single soundness axiom? Do P1 and P2 imply P3 for any natural class of encoders?
3. For Anthropic's Llama-3 SAE ($m \approx 65$ k, $n < 10$ k), the null-encoder upper bound from §3.4 is $\sqrt{2 \log 6.5} \approx 2.16$. How does this compare to their reported MCC-type metrics? Can this comparison be made during the rebuttal?
4. What is the closed-form (or asymptotic) expression for R^2 under nonlinear overcomplete E6?
5. Will the §C survey's search queries, inclusion/exclusion criteria, and inter-rater reliability be published in the appendix?
6. When will the stress-test suite and practitioner checklist be released? Is an anonymous.4open.science link available now?

Q6 rating: 6: Weak Accept: Technically solid paper, with no major concerns with respect to provided evidence, resources, reproducibility, and ethical considerations.
Q7 justification:
Treating metric failure as population-level misspecification is novel and relevant. The Fig 1 caption vs Tab 3/Fig 4 contradiction on R^2's P3 status directly affects a headline claim, dropping my score from 7 to 6 pending clarification. Resolving the contradiction + adding the Anthropic SAE case study → 8; contradiction only → 7; unresolved → 5.

Q8 confidence: 4: Quite confident. I tried to check the important points carefully. It is unlikely, though conceivable, that I missed some aspects that could otherwise have impacted my evaluation. I am familiar with the research topic and most of the related work.
Q9 confirmation: I have read the UAI reviewing instructions and certify that I comply with them.
Q10 ethics:
None.

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Official Review of Submission1105 by Reviewer wGET

Official Review by Reviewer wGET 08 Apr 2026 at 21:54 (modified: 24 Apr 2026 at 04:15)

Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer wGET, Authors Revisions (/revisions?id=dBY644tkXg)

Q1 summary:

This work studies commonly used evaluation metrics in identifiable representation learning and characterizes several failure scenarios, including correlated latent representations, redundancy in ground truth factors due to deterministic (causal) relationships, over-parameterized encoders, and low representation-to-sample ratios. For each scenario, the authors present the desired property, synthetic empirical examples, and in some cases provide theoretical justification.

Q2 novelty: 3: Good: The paper makes non-trivial advances over the current state-of-the-art.
Q2 correctness: 3: Good: The paper appears to be technically sound, but I have not carefully checked the details.
Q2 evidence based support: 3: Good: the main claims are supported by convincing evidence (adequate experimental evaluation, proofs, code, references, assumptions).
Q2 reproducibility: 3: Good: key resources are available and key details are sufficiently well-described for competent researchers to confidently reproduce the main results.
Q2 writing clarity: 4: Excellent: The paper is well-organized and clearly written.
Q3 strengths:

1. This work studies an important yet often overlooked problem in identifiable representation learning: whether commonly used metrics faithfully reflect identifiability. Analyzing failure cases from a novel perspective and characterizing them offers useful insights that could benefit the community.
2. The writing is great. The whole paper is well-structured and well-explained, with clear notations, illustrative examples, and highlighted takeaway.
3. The empirical and theoretical analysis is thorough, generally well supports the authors' claim and intuition.

Q4 weaknesses:

1. I am not fully convinced that the central question of section 3.2 (recover lossless part only) aligns with the goal of identifiable representation learning. For example, CRL aims to recover the whole set of causal variables, including the graphical structure. Even in deterministic settings, latent variables can still carry semantic meaning. From this perspective, it may be preferable for a metric to penalize missing variables more explicitly. It would be nice to explain more about the relationship between these two goals.
2. As the author also discussed in the final section, all the empirical analysis is based on synthetic $\hat{Z}$, which is indeed more straightforward but also limits the practical impact of the findings, leaving a gap to the real training model.
3. This work mainly focused the potential problems of existing metrics and provides concrete failure cases. However, except for a practical checklist, no improved metric or clear design principles for better evaluation are proposed. Also, the checklist assumes access to ground-truth information such as latent dimensionality d, which may not be available in practice. This creates a circular situation: if d is known, model design could already enforce this structure, reducing the need for such evaluation guidelines.

Q5 comments:

The caption of Figure 3 might mismatch the content, and D_f should be the middle one based on the context in the main text, and D_F is missing. Also, the D1-D4 used in the title is inconsistent with the notation used in the main text.

Q6 rating: 7: Accept: Technically solid paper, with convincing evidence for claims, innovative aspects, good resources and reproducibility, no unaddressed ethical considerations, and the potential for impact in the field.
Q7 justification:
This work provides a thorough, well-explained analysis of limitations in widely used metrics for identifiable representation learning, provide valuable insights for future research in this area.

Q8 confidence: 4: Quite confident. I tried to check the important points carefully. It is unlikely, though conceivable, that I missed some aspects that could otherwise have impacted my evaluation. I am familiar with the research topic and most of the related work.
Q9 confirmation: I have read the UAI reviewing instructions and certify that I comply with them.
Q10 ethics:
no

Add: **Official Comment** **Rebuttal**

Official Review of Submission1105 by Reviewer hjjj

Official Review by Reviewer hjjj 01 Apr 2026 at 23:10 (modified: 24 Apr 2026 at 04:15) Program Chairs, Senior Area Chairs, Area Chairs, Reviewers Submitted, Reviewer hjjj, Authors
Revisions (/revisions?id=OhAGrbST5R)

Q1 summary:

The paper asks whether standard identifiability metrics (MCC, R^2, DCI) actually measure recovery of ground-truth factors. It shows they are trustworthy only under specific structural assumptions and otherwise can give systematic error. Its main contributions are a taxonomy of DGP vs. encoder assumptions, a characterization of each metric's validity domain, a reproducible stress-test suite, and more importantly, the evaluation guideline for practitioners distilled from the findings.

Q2 novelty: 3: Good: The paper makes non-trivial advances over the current state-of-the-art.
Q2 correctness: 3: Good: The paper appears to be technically sound, but I have not carefully checked the details.
Q2 evidence based support: 3: Good: the main claims are supported by convincing evidence (adequate experimental evaluation, proofs, code, references, assumptions).
Q2 reproducibility: 2: Fair: key resources are unavailable but key details are sufficiently well-described for an expert to confidently reproduce the main results.
Q2 writing clarity: 3: Good: The paper is well organized but the presentation could be improved.
Q3 strengths:

1. **The paper studies an important and under-explored problem.** It examines when commonly used identifiability metrics can be interpreted as meaningful evidence, a question of clear relevance to disentanglement, nonlinear ICA, and causal representation learning. This issue is central in practice, yet is often left implicit.
2. **The paper gives a clear and useful account of when and why these metrics fail.** The proposed taxonomy cleanly separates assumptions on the data-generating process from those on encoder geometry, and the analysis makes the failure modes of existing metrics precise. The contribution is not merely critical; it provides a structured understanding of metric validity.
3. **The empirical study is well aligned with the paper's objective.** The controlled constructions isolate metric behavior from optimization and implementation effects, which is the right design for the question being studied. The experiments are sufficient to reveal the main phenomena clearly and convincingly.
4. **The paper is well organized and easy to follow.** The presentation is clear throughout, with examples and takeaways chosen effectively. The appendix section with practical guidance is particularly useful, as it translates the main findings into concrete advice for researchers using these metrics.

Q4 weaknesses:

1. **The paper's practical scope is still limited by the absence of learned-encoder evidence.** The central analysis is intentionally conducted with synthetic encoders to isolate metric behavior, and the paper itself notes that studying different families of learned encoders is left to future work; as a result, it remains unclear when the proposed geometries, especially the overcomplete ones, arise under realistic training dynamics.

2. **The overcomplete encoder taxonomy remains somewhat stylized, even in its most complex cases.** The paper's conclusions for distributed overcomplete representations rely on controlled constructions such as disjoint E8 subsets, which are useful for isolating metric failures but still leave some distance to more general mixed or distributed code geometries that may occur in practice.
3. **The paper is stronger as a diagnosis of current metrics than as a practical resolution of the evaluation problem.** It convincingly shows that no existing metric is reliable across all common settings, but the practical outcome is mainly a taxonomy, stress-test suite, and cautionary guidance, rather than a concrete evaluation protocol or metric that clearly resolves the problem in realistic pipelines.

Q5 comments:

Major

In general, this paper has well-studied when and why existing prevalent metrics for evaluating disentanglement could fail, revealing useful insights especially via a clear taxonomy that separates DGP effects from encoder geometry. While the authors have carefully considered different cases, e.g., E4-E8, it still falls short on idealized modeling and lack of complex empirical evidence, as also acknowledged in the limitation section. Specifically, E8 is the most complicated case for overcompleteness ($m > d$), yet still considering disjoint indices among latents and not modeling case where single latent may also correspond to other factors (many-to-many). More importantly, one may further ask whether these categories are necessary without sufficient justification via empirical observations at least on benchmark data, e.g., Shapes3D. It thus makes one wonder in what condition or inductive bias should we expect correct behaviours as characterized by these cases. This paper could be much improved by such evidence.

Minor

- The notation n is repeatedly used in both data generative process and number of samples. Please clarify.
- The Proposition 1 seems not really generalisable but more of like an example showing failure case of MCC. It's still valuable however I would suggest re-framing the claim. At least one could write it in an improved form by clearly stating the assumption and quantitative relation between ρ and $MCC - P$.
- Controlled settings are at the core and throughout this paper which are highly dependent upon implementations of disentanglement metrics (MCC, DCI-D, R^2). While the authors promise to release to public upon acceptance, this weakens reproducibility and confidence.

Questions

- Does there exist any dimensionality reduction approaches to properly handle the $m > d$ case? If so, what are the assumptions?
- Can we properly decide m by estimating intrinsic dimension $\hat{d}$ given data observations?

Q6 rating: 6: Weak Accept: Technically solid paper, with no major concerns with respect to provided evidence, resources, reproducibility, and ethical considerations.
Q7 justification:
This paper studies an important yet less-explored question in disentangled representation learning, providing detailed analyses via contributing taxonomy which leads to insights for both better evaluation methods and practitioners. The reason I did not recommend a higher score is due to the limitation on lack of empirical evidence to justify considered encoder geometry cases.

Q8 confidence: 3: Somewhat confident, but there's a chance I missed some aspects. I did not carefully check some of the details, e.g. novelty, proof of a theorem, experimental design, or statistical validity of conclusions. I am somewhat familiar with the topic but may not know all related work.
Q9 confirmation: I have read the UAI reviewing instructions and certify that I comply with them.

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